

Problem-Solving Tutorial 3

Spin 1/2 Systems: One and Two Particles

The problem-solving tutorials will have a mix of group discussion questions and questions for individual (or small group study). You are welcome to work with your friends and neighbours, and please ask the tutor questions. We want these sessions to be as interactive as possible.

1. **[Discussion]** Please look at the statements, and identify why they are wrong or “not quite right”:
 - (a) For an angular momentum state, $|jm\rangle$ the magnitude of the total angular momentum is $j(j+1)\hbar$.
 - (b) Considering that in the generalized uncertainty relations the uncertainties in L_x, L_y are related to $[\hat{L}_x, \hat{L}_y]$, I have concluded that for an atomic orbital angular momentum state $|l0\rangle$, with zero L_z component, there is zero uncertainty in the measurements of either L_x and L_y .
 - (c) The matrices for $\hat{J}_x, \hat{J}_y$ and $\hat{J}_z$ are given below. Defining an eigenvector $\mathbf{c}^T = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$, then if $\hat{J}_x \mathbf{c} = -\mathbf{c}$ then we can infer that the eigenvalue is equal to -1.
2. **[Individual]** For a $J = 1/2$ system, work out the matrix representation of $\hat{J}_x$ and J_y in the basis $|jm\rangle$.
3. **[Individual]** The state vector $|\phi\rangle$ given by:

$$|\phi\rangle = c_1|11\rangle + c_0|10\rangle + c_{-1}|1-1\rangle$$

is a linear superposition of the three vectors $|jm_j\rangle$ for which $j = 1$ and $m_j = 1, 0$ and -1 . For what values of c_1, c_0 and c_{-1} is $|\phi\rangle$ an eigenvector of $\hat{J}_x$ having eigenvalue 0? Your answer should ensure that $|\phi\rangle$ is normalised to unity.

4. **[Individual]** A spin- $\frac{1}{2}$ particle is acted on by a magnetic field $\mathbf{B}$ acting along the z -axis, so $\hat{H} = -\mu_B |\mathbf{B}| \sigma_z$, where μ_B is the Bohr magneton.

At $t = 0$, the state of the system is given by $|\psi(0)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ in this state the x -component of spin angular momentum $\hat{S}_x = \frac{1}{2}\hbar\sigma_x$ has the precisely defined value $+\frac{1}{2}\hbar$. Obtain a formula for the time t at which the x -component of spin angular momentum first has the value $-\frac{1}{2}\hbar$. (At this time t , the state vector is $|\psi(t)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ multiplied by an appropriate phase factor.)

5. **[Special Question — there will be no solutions until after the assessed problem sheet is submitted. This is to practice the learning from week 8 in class with colleagues.]** In these questions, we are going to practice using C-G Coefficients by expressing systems in the particle basis or the total angular momentum basis.

- (a) Express $|j_1 = \frac{3}{2}, m_1 = \frac{1}{2}\rangle |j_2 = \frac{1}{2}, m_2 = \frac{1}{2}\rangle$ in the total angular momentum case.
- (b) Express $|J = \frac{5}{2}, M = \frac{3}{2}; j_1 = 2, j_2 = \frac{1}{2}\rangle$ in the particle basis.
- (c) For the state $|J = \frac{5}{2}, M = \frac{3}{2}; j_1 = 2, j_2 = \frac{1}{2}\rangle$, work out $\langle \hat{S}_{x1} \rangle$.

The operator matrices for a system with angular momentum quantum number $j = 1$ are given by:

$$\hat{J}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{J}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \hat{J}_z = \hbar \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

You may assume without proof that:

$$\begin{aligned} \hat{J}_+ |jm_j\rangle &= \hbar [j(j+1) - m_j(m_j+1)]^{1/2} |jm_j+1\rangle \\ \hat{J}_- |jm_j\rangle &= \hbar [j(j+1) - m_j(m_j-1)]^{1/2} |jm_j-1\rangle. \end{aligned}$$

CLEBSCH-GORDAN COEFFICIENTS

$$j_1 = \frac{1}{2}, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = 1$$

m_1, m_2	j	1
$\frac{1}{2}, \frac{1}{2}$	$\frac{1}{2}$	1

$$m = -1$$

m_1, m_2	j	1
$-\frac{1}{2}, -\frac{1}{2}$	$-\frac{1}{2}$	1

$$m = 0$$

m_1, m_2	j	1	0
$\frac{1}{2}, -\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$-\frac{1}{2}, \frac{1}{2}$	$-\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$j_1 = 1, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = \frac{3}{2}$$

m_1, m_2	j	$\frac{3}{2}$	1
$1, \frac{1}{2}$	$\frac{1}{2}$	1	1

$$m = \frac{1}{2}$$

m_1, m_2	j	$\frac{3}{2}$	$\frac{1}{2}$
$1, -\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{2}{3}}$
$0, \frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$

For brevity, solutions with $M < 0$ and $j_1 < j_2$ are omitted. They may be calculated using the simple relations

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M \rangle = (-1)^{J-j_1-j_2} \langle j_1, j_2; -m_1, -m_2 \mid j_1, j_2; J, -M \rangle,$$

and

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M \rangle = (-1)^{J-j_1-j_2} \langle j_2, j_1; m_2, m_1 \mid j_2, j_1; J, M \rangle.$$

$$j_1 = \frac{3}{2}, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = 2$$

m_1, m_2	j	2	1
$\frac{3}{2}, \frac{1}{2}$	$\frac{1}{2}$	1	1

$$m = 1$$

m_1, m_2	j	2	1
$\frac{3}{2}, -\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$
$\frac{1}{2}, \frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$-\frac{1}{2}$

$$m = 0$$

m_1, m_2	j	2	1
$\frac{1}{2}, -\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$-\frac{1}{2}, \frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$j_1 = 1, j_2 = 1 \quad [\text{edit}]$$

$$m = 2$$

m_1, m_2	j	2	1
$1, 1$	1	1	1

$$m = 1$$

m_1, m_2	j	2	1
$1, 0$	$\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$0, 1$	$\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$m = 0$$

m_1, m_2	j	2	1	0
$1, -1$	$\frac{1}{6}$	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$
$0, 0$	$\frac{2}{3}$	$\sqrt{\frac{2}{3}}$	0	$-\sqrt{\frac{1}{3}}$
$-1, 1$	$\frac{1}{6}$	$\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$